

Zero

material environmental incidents in FY20

Environmental Performance Standards

Our Sustainability Performance Standards contain eight key business risk areas related to environment: air quality, biodiversity, rehabilitation and mine closure, resource efficiency and emissions, tailings storage facilities, waste, waste rock and ore, and water.

All our sites and workplaces are required to meet the Standards which are audited on a regular basis. Details of the performance of the eight business risk areas are available on our website¹.

Environmental data (water, air emissions and energy) at our operations is collated and verified by third party auditors Greenbase. FY20 data can be found on p.49-50. They are reported monthly and measures have been set to establish baseline improvements year on year.

Climate-related risks, emissions and energy

Evolution acknowledges that climate-related risks, both transitional and physical, have the potential to impact our business, communities and the environment. We are committed to understanding and proactively managing these risks.

Sound risk management practices and strategic planning are integrated across all areas of our business, leveraging off technology to ensure projects return long term value. Comprehensive Board-level governance and oversight are applied.

In FY20, as a first step towards the TCFD framework, emerging climate-related risks were identified and included in the Risk Register.

Climate-related risks are actively reported, and targets have been developed for FY21 to build on opportunities, improve energy efficiency and water security, prepare for extreme weather and health events, (including pandemic and smoke impacts from fires) and adopt responsible water management practices. Specific targets have also been captured in the Balanced Business Scorecard, including implementing community plans, year-on-year reduction of

CO₂-e emissions per tonne of material mined, and year-on-year reduction of raw water demand per dry tonne milled based on historical data. The case studies for water efficiency at Cowal, extreme weather at Mt Carlton and the management of COVID-19 demonstrate our ongoing commitment and mitigation of climate-related risk.

The climate-related physical risks identified as applicable to our business are as follows:

- **Energy and emissions:** We keep informed of changing regulations, including policy, codes and principles to help manage transition risk. We engage with our community and stakeholders to ensure we are operating with a holistic mindset. We remain agile in response to changing markets and explore innovative technology including renewables to improve our resilience to resource financial and supply uncertainty. We aim to contribute positively to local, regional and national sustainability efforts 
- **Water security:** Production is reliant on the availability of water. In the short term, the Group is adapting to a changing water security environment by working towards reducing demand and reusing a greater portion of water. In the medium to long term, we are investing in water recycling and reuse strategies to improve efficiencies 
- **Extreme weather events:** We ensure we minimise the impact of extreme weather events on our operations through business continuity planning. This includes the consideration of potential climate impacts in the design and construction of new and upgraded assets and alternate supply chains 
- **Extreme health events:** the events of COVID-19 this year have impacted globally and have highlighted the need to act early and collectively to mitigate loss, both of life and financial. We also recognise that we must remain prepared to manage these events and support the communities in which we operate with their recovery efforts. We have integrated this into our scenario plans, supply chain reviews and we have stress tested our balance sheet and our financial capacity 

1. <https://www.evolutionmining.com.au/environment>

CASE STUDY: Pumped hydro renewable energy at Mt Rawdon

We are progressing with the feasibility project to integrate renewable energy infrastructure at the Mt Rawdon Operation. The pumped hydro scheme would entail the use of the open pit void as the lower reservoir and the construction of an upper reservoir. There is potential for long-term benefits to the region and economy should the project become a reality.

CASE STUDY: Energy efficiency initiatives at Cowal

During FY20, options were identified to reduce energy use and emissions at the Cowal Operation. Through remote monitoring and analysis of the processing plant using business intelligence software MillROC, processing efficiency was optimised and electricity and inputs demand in the processing plant were reduced. The MillROC efficiency program also identified opportunities to reduce wear and grinding media requirements to achieve optimum grind in the milling circuit for improved gold recovery.

Emissions reporting

Each year, we submit annual reports¹ for Australia's National Pollutant Inventory (NPI) and the National Greenhouse and Energy Reporting Act 2007 (NGER Act) to estimate greenhouse gas (GHG) emissions and energy use at our Australian operations.

In FY20, we developed baseline data and analysis for each operation to understand the source of peak energy use and emissions. We will continue this work in FY21 through a newly formed Resource Efficiency Working Group that aims to implement strategies to reduce climate-related risk through analysis of past performance and sharing of innovation and technology initiatives.

Emissions performance

Overall, the Group's total GHG emissions increased 3% compared to FY19, with a 2% increase in Scope 1 emissions and a 4% increase in Scope 2 emissions. The increase is largely a result of an increase in activity at Mt Carlton and an increase in electricity purchased at Cowal. In line with the increase in emissions, energy consumed also increased by 3%.

At our Cowal Operation total Scope 1 emissions were stable compared to FY19 with diesel use (predominantly for mining) decreasing by 2% and LPG (used in processing) increasing by 26%. Electricity use increased by 6% in line with a 7% increase in ore processed. However Scope 2 emissions increased by 5% due to NSW grid factors increasing. Electricity use increased by 6% in line with a 7% increase in ore processed.

At our Mt Carlton Operation, emissions increased in FY20 due to underground developments and process plant upgrades. This resulted in Scope 1 and Scope 2 emissions increasing in comparison to production metrics. These developments, along with the construction of a new TSF, support life of mine (LOM) ore processing and long-term storage of by-products. The use of enhanced water evaporation powered by diesel has been an integral part of a new water management system and has led to Scope 1 emission increases.

Water management

We recognise the importance of water to our business and our communities and take a proactive approach to responsible water management and security. Our standards aspire to reduce the demand for fresh water and the overall requirement for water in our operations. The water management performance standard ensures that our operations effectively manage water, including process water, stormwater, discharges and dewatering activities.

FY20 achievements

- 85% water utilised for operations was recirculated for reuse onsite
- 11% increase in water reuse between FY19 and FY20
- 0.74kL average water demand per tonne of ore milled at all operations

Our total water withdrawn increased by 21% in FY20. The increase in water withdrawn was due to mine dewatering at all operations with increases at Mt Carlton and Mungari.

Total water recycled and reused increased by 11% between FY19 and FY20. We use recycled water primarily to process ore and in other activities such as paste fill, irrigation, dust suppression and construction.

Operations prepare for seasonal variations in water flow and maintain routine dewatering activities to satisfy water licence conditions. Probabilistic site water balance models are used to predict water flow and requirements during droughts and stormwater flows at operations.

In FY20, one uncontrolled release of water of more than 10kL occurred. Cracow had a 11.7kL release of mine affected water which was contained with no environmental harm or enforcement action.

1. <https://evolutionmining.com.au/environment>

In FY21, there is a focus on reducing the volume of fresh water used and increasing the portion of water reuse for all operations.

CASE STUDY: Cowal water usage

During summer 2019/20, the Lachlan catchment of NSW in which our Cowal Operation is located faced Level 3 water restrictions due to prolonged drought conditions. Although our mining operations were not materially impacted by the restrictions, agricultural and municipal areas in the region were. Evolution responded by adopting a regional mindset to identify water saving measures at the Cowal Operation. After polling 300+ employees on ways of decreasing raw water use and demand, our Cowal Water Strategy Group introduced a series of initiatives that included:

- Increasing reuse of mine water from current 47% average
- Increasing site water storage capacity from 725ML to 1080ML
- Reducing the operation's reliance on fresh water by increasing saline groundwater use from 8% (2ML) per day to 16-20% (4ML) per day
- Reducing water demand for the operation from the current 24ML per day

Waste management

The major waste streams in mining are tailings and waste rock. Minor waste streams include industrial waste, organic and inorganic waste. The management of waste is supported by group wide policies and performance standards and local operating procedures.

All waste streams are monitored through regular inspections and waste tracking.

Operations use specialist, licenced waste management service providers and tracking arrangements for the approved and safe disposal of transfers of obsolete or used hazardous materials. Generally, chemicals are consumed in the process. Regulated waste streams are transferred off-site for industrial re-refining (for reuse) or converted into energy.

Operations seek to reduce impact to our communities and the environment through elimination, reduction, recycling and reuse of waste streams in favour of disposal.

- **Elimination:** waste converted to energy (pre-feasibility studies in progress)
- **Reduction:** 43% reduction in hazardous chemical use achieved at Cracow and Mt Rawdon has significantly increased their haul truck hours reducing the environmental footprint

- **Reuse:** Waste rock is used for the construction of TSF embankments at most of our operations forming Integrated Waste Landforms (IWL) reducing our environmental footprint
- **Recycle:** investigating sustainable recycling of tyres and PPE clothing options with community partners

43%

reduction in hazardous chemical use at Cracow

CASE STUDY: Reduction in reagents use, Cracow

In FY20, a hazardous chemical reduction project at the Cracow Operation was completed. This resulted in a 43% reduction in the consumption of critical hazardous chemicals in the process plant - hydrogen peroxide consumption was cut by 75% and cyanide consumption by 20% relative to FY19. The resulted in a reduction in exposure to the chemicals within the local environment of the operation and a reduction in transport requirements and tailings disposal. This change was completed without any negative impact to recovery. In FY21 the learnings from this will be applied to other operations to increase the benefits to communities and the environment.

Tailings management

We maintain 11 TSFs in Australia, eight of which are active, one under construction and two decommissioned.

In producing 746,463oz gold, 15.06Mt of ore was processed and deposited into storages.

More than 700kt of approximately 1.5Mt tailings were recovered for reuse at the Mungari Operation.

Our tailings governance and assurance frameworks incorporate the six key components of the Tailings Governance Framework as outlined by the International Council on Mining and Metals (ICMM).

Our Tailings Storage Facility Governance Committee provides effective oversight of our policies, standards and practices with respect to tailings management.

A Group Head of Technical Storage Management was appointed in FY20 to oversee such operations and gain greater insight into the risks and mitigation potential associated with TSF management.

Performance data

Safety

Safety performance

	FY20 ¹	FY19	FY18	FY17	FY16	FY15
Employee fatalities	0	0	0	0	0	0
Contractor fatalities	0	0	0	0	0	0
Total Recordable Incident Frequency (TRIF)	6.76	8.31	5.49	7.96	9.70	9.60
Lost Time Injury Frequency (LTIF)	2.07	1.75	0.50	0.40	1.80	1.00
Significant Safety Occurrence Frequency (SSOF)	n/a	n/a	3.20	4.95	5.9	11.61 [^]
Safety Incident Frequency (SIF)*	14.84	15.97				

All classifications above include contractors

*In FY19 Evolution changed the definition of significant safety occurrence and renamed to significant incidents. The definition change has resulted in more incidents being classified as significant incidents

[^]FY15 figures are for Evolution operated assets. In the FY16 annual report Mungari and Cowal FY15 figures had been added for comparative purposes (operated by previous owner)

Emergency Rescue Teams (ERT)

	Number of ERT members				Community responses			
Cowal	28				0			
Cracow	25				1			
Mt Carlton	32				2			
Mungari	28				1			
Mt Rawdon	23				0			
Total	136				4			

Diversity

Profile FY20 ²	Full time		Part time		Fixed term		Casual		Employees		Contractors	Labour hire	Workforce
	M	F	M	F	M	F	M	F	Total	Total	Total	Total	Total
	1,154	202	1	16	38	12	16	9	1,209	239	554	97	2,099

* Service providers for specifically required roles ^ Temporary replacements for current Evolution positions

Levels FY20	Board		Senior Leaders		Manager / Superintendents		Professionals / Supervisors		Operations	
	M	F	M	F	M	F	M	F	M	F
	6	2	16	2	132	28	211	55	850	154

New Employees FY20	Age group <36		Age group 36-55		Age group >55		Total
	M	F	M	F	M	F	
	116	62	100	34	22	4	338

Turnover FY20	Age group <36		Age group 36-55		Age group >55		Total
	M	F	M	F	M	F	
	25%	13%	47%	9%	7%	0%	11%

Turnover result for EVN total is voluntary permanent only 12 month moving average result. The split by age bracket and gender is the % of all terminations (summing up to 100%), rather than turnover rates for each grouping

Site profile FY20	Corporate		Cowal		Mungari		Mt Carlton		Mt Rawdon		Cracow		Total		
	M	F	M	F	M	F	M	F	M	F	M	F	M	F	All
Employees	84	53	358	63	221	61	152	17	182	25	212	20	1209	239	1,448

State profile FY20	NSW		WA		QLD	
	M	F	M	F	M	F
Employees	442	116	221	61	546	62

Developing our approach to diversity and inclusion	FY20 target	FY20 Actual	FY19
Female graduate intake	50%	62%	50%
Female summer vacation interns	50%	64%	39%
Indigenous and First Nation employees ³	7%	7%	4.8%

- Includes Red Lake Operation TRIF from 1 April 2020.
- Number of employees in Australia as at 30 June 2020

- FY20 number includes Red Lake Operation

Environment

Environmental incidents

Incident level	FY20	FY19	FY18	FY17
Catastrophic	0	0	0	0
Major	0	0	0	0
Moderate	8	9	7	9

Energy

Energy consumption (GJ)	FY20	FY19	FY18	FY17	FY16
Total	4,102,171	3,986,905	4,075,493	4,402,695	4,415,040
Energy intensity (ore processed - GJ/tonne)*					
Total	0.270	0.280	0.291	0.272	0.273

*Ore processed does not include our financial stake in Ernest Henry

Emissions

Total direct and indirect emissions	FY20	FY19	FY18	FY17	FY16
Greenhouse gas emissions Scope 1 (t CO ₂ -e) ¹	157,857	155,085	159,061	167,734	157,584
Greenhouse gas emissions Scope 2 (t CO ₂ -e) ²	398,187	383,449	394,144	430,993	472,257
Total of Scope 1 and Scope 2 (t CO ₂ -e)	556,044	538,533	553,205	598,727	629,841
Sulphur oxide SO _x (kg)	932	916	926	1,002	936
Nitrous oxide NO _x (kg)	1,492,143	1,397,676	1,395,277	1,521,718	1,627,006
Carbon Monoxide (CO) (kg)	722,095	719,005	701,930	886,918	788,636
Particulate matter < 10 µm (kg)	4,730,467	4,260,114	4,323,757	5,163,574	5,585,918
Particulate matter < 2.5 µm (kg)	96,844	88,471	86,683	95,254	104,208
Total volatile organic compounds (VOC) (kg)	100,959	90,036	86,380	97,958	110,103
Emissions of lead and lead compounds	688	577	639	812	1,147
Emissions of mercury and mercury compounds	1.29	1.55	1.48	2.25	3.64
Emissions intensity (ore processed - t CO ₂ -e/tonne)					
Total	0.037	0.038	0.039	0.037	0.039

The energy and emissions boundary is based on operational control as defined by the National Greenhouse and Energy Reporting (NGER) Act 2007. The applied global warming potential (GWP) rates and emission factors are based on the NGER Act (2007) and the National Pollutant Inventory

1. Scope 1 refers to emissions produced directly by operations, primarily resulting from combustion of various fuels and includes CO₂-equivalent values for greenhouse gases such as CH₄, N₂O and SF₆
2. Scope 2 refers to indirect emissions resulting from the import of electricity from external parties; commonly the electricity grid

Water withdrawal

Water withdrawal Surface (ML) water	FY20	FY19	FY18
Surface water (ML)	4,948	3,506	2,077
Groundwater - mine dewatering (ML)	3,034	3,034	3,212
Groundwater - borefields (ML)	826	1,589	1,476
Rainwater (ML)	2,127	1,000	1,856
Municipal water (ML)	74	66	59
Total water withdrawal	11,087	9,194	8,680
Reused (ML)	9,475	8,545	7,018
% Total reused	85%	93%	81%
Water intensity (ore processed - kL/tonne)	0.74	0.65	0.62

Water discharge

Water discharge (ML)	Surface water	Sewers that lead to water	Land - Dust suppression	Land - Irrigation	Groundwater	Treatment Facility	Total
Total	1,726	0	915	0	0	0	2,641

Environment

Waste				
Mineral waste	Waste material mined (kt)	Solids in tailings (kt)	Total ore processed (kt)	Explosives (t)*
Total*	31,563	15,011	15,065	11,414

* The reporting period for liquid fossil fuels, lubricants and explosives is July 2019 to June 2020 and reported as part of the National Pollutant Inventory.

Tailings					
Asset	Cowal	Cracow	Mt Carlton	Mt Rawdon	Mungari
Active TSF	2	2	1	1	1
Inactive TSF	0	3	0	0	0
Construction Type	Upstream / IWL	Upstream	Downstream, HDPE Lined	Upstream / IWL	Upstream / IWL

Non-mineral waste							
Tonnes	Off-site Landfill	On-site landfill	Recycling / Reuse	Tyres disposed of on-site	Tyres incinerated	Composted or bio-remediated	Other - Septic waste
Total	3,267	1,573	4,814	9	40	6,027	2,285

Acid-generating seepage					
Asset	Cowal	Cracow	Mt Carlton	Mt Rawdon	Mungari
Predicted to occur	-	-	-	-	-
Actively mitigated	Yes	Yes	Yes	Yes	Yes
Under treatment or remediation	-	Yes	-	-	Yes

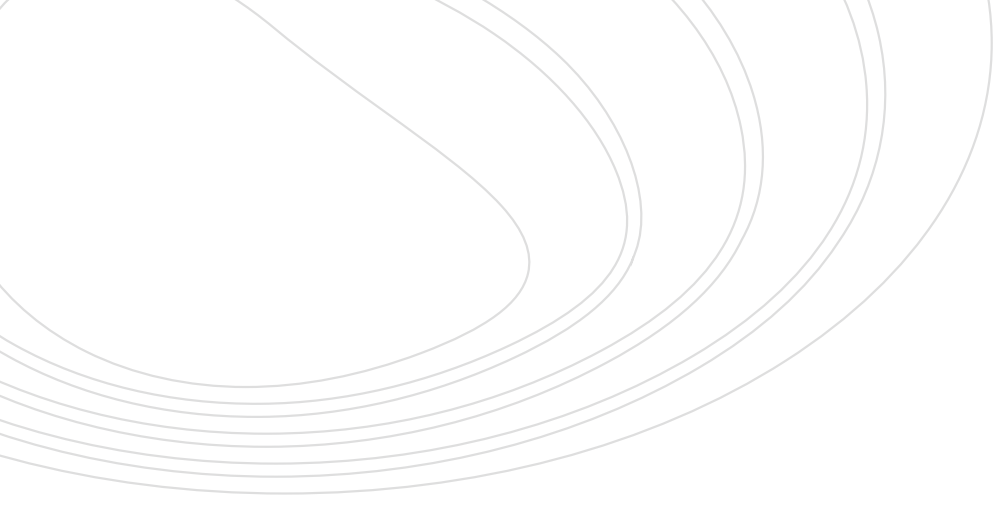
Rehabilitation and closure			
Land management (ha)	Land disturbed	Land rehabilitated	Sites with protected conservation status
FY20 only	213	35	1
Total	2,193	188	1

Environmental compliance	
Total volume of significant spills	13
Monetary value of significant fines (\$A)	0

Glossary and acronyms

“A” rating	Rating credibility used in the DJSI review. The lowest rating of “CCC” to the highest rating of “AAA”
A\$	Australian dollars
ALO	Act Like an Owner. An internal ongoing award to reward our employees for their supportive behaviour
AMD	Acid mine drainage. When sulphide minerals (predominantly pyrite) are exposed to air, which allows them to oxidise and break down
B	Billion. The number equivalent to the product of a thousand and a million
CMT	Crisis management team. The CMT provides support through management of crisis level issues
CN	Cyanide. A chemical compound used in the extraction of gold and silver
COVID-19	Severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) is the strain of Novel coronavirus that causes coronavirus disease 2019. A mild to severe respiratory illness that is caused by a coronavirus and is transmitted chiefly by contact with infectious material (such as respiratory droplets) or with objects or surfaces contaminated by the causative virus
CSA	Corporate Sustainability Assessment. A scoring methodology that companies and investors can review on a company's ESG
CSIRO	Commonwealth Scientific and Industrial Research Organisation. An Australian government agency responsible for scientific research
Dewatering	The act of taking water from an operating mine
DJSI	Dow Jones Sustainability Indices. These are a family of indices evaluating the sustainability performance of thousands of companies globally
EAP	Employee assistance program. Program available to employees and their families to use to assist with their health and wellbeing
ERT	Emergency Response Team. Teams built at each operation to support both our operations and assist communities through significant incidents or threatening situations
ESG	Environmental, Social and Governance. The three key factors when evaluating the sustainability and ethical impact of an investment in a company or country
ESS	Employee Share Scheme. A scheme introduced by Evolution Mining 6 years ago which supports the issuing of shares to our full and part-time employees to ensure they share in Evolution's success
FY20 / FY21	FY meaning financial year. FY20 would then be the period from July 2019 to end of June 2020
GHG	Greenhouse Gas. Compound gases that trap heat or longwave radiation in the atmosphere
ICMM	International Council on Mining and Metals. An international organisation whose purpose bringing together a safe, fair and sustainable mining and metals industry
ISS ESG	Institutional Shareholder Services (ISS). ISS ESG is a business that provides corporate and company ESG research and ratings
IWL	Integrated waste landform. A simple definition is a tailings storage facility that is located inside waste rock storage
JT	Johnathan Thurston. He is an Australian former professional rugby league footballer who has established an academy to provide employment initiatives and training
kL	Kilolitre. Measurement equivalent to 1,000 litres
LGBTI+ community	Loosely defined grouping of people who are lesbian, gay, bisexual, transgender/gender diverse, intersex and other minorities
LOD	Line of Defence
LoKal	Name given to a local community initiative in Kalgoorlie

M	Million. Number equivalent to the product of a thousand and a thousand
MillROC	Milling Remote Optimisation Consulting & Coaching. Software produced by Orway IQ which is a cloud-based reporting of all plant data related to circuit performance and optimisation
ML	Megalitre. Equal to one million litres
MSA	Modern Slavery Act. The Commonwealth Modern Slavery Act 2018 (the Act) established Australia's national Modern Slavery Reporting Requirement (reporting requirement). The reporting requirement entered into force on 1 January 2019. The reporting requirement aims to support the Australian business community to identify and address their modern slavery risks and maintain responsible and transparent supply chains
MSCI	Morgan Stanley Capital International. It is an investment research firm
NGER	National Greenhouse and Energy Reporting. A national framework for reporting and disseminating company information and greenhouse gas emissions, energy production and energy consumption
NGOs	Non-governmental organisation. A non-profit, citizen-based group that functions independently of government
NMD	Neutral mine drainage. In some instances, the acidity produced by sulphide oxidation can be neutralised in the presence of carbonate minerals
NPI	National Pollutant Inventory. The NPI provides the community, industry and government with free information about substance emissions in Australia
PAF	Potentially Acid Forming. Classification of a rock when tested if it has the potential to generate acid as a result of a metal mining activity
PPE	Personal protective equipment. Anything used or worn on our employees to minimise risk to their health and safety
S&P Global	Company that provides data, research, news and analytics to customers including institutional investors and corporations
SAM	Title for the Corporate Sustainability Assessment. SAM refers to historic naming when the CSA was hosted by RobecoSAM AG. It is now transferred to S&P Global Switzerland SA and known as the SAM Corporate Sustainability Assessment
Scope 1	Category of greenhouse gas emissions. Scope 1 is sometimes referred to as direct emissions and refers to emissions released to the atmosphere as a direct result of an activity
Scope 2	Category of greenhouse gas emissions. Scope 2 refers to emissions released to the atmosphere from the indirect consumption of an energy commodity
SD	Saline drainage. This is saline and metal-rich drainage that has been produced by the oxidation of metal sulphides that do not generate net acidity
t	tonnes
TCFD	Task Force on Climate-related Financial Disclosures. An organisation that was established in December 2015 with the goal of developing a set of voluntary climate-related financial risk disclosures which may be adopted by companies
TRIF	Total Recordable Injury Frequency. Usually forms part of the acronym TRIFR and refers to the number of fatalities, lost time injuries, alternate work, and other injuries requiring medical treatment per million hours worked
TSF	Tailings storage facility. A facility designed to safely store left over mined minerals
UN SDGs	United Nations Sustainable Development Goals. These are global goals adopted by all United Nations Member States as a universal call to action to end poverty, protect the planet and ensure that all people enjoy peace and prosperity by 2030
WORK 180	A recruitment site showing Australian employers who support women in the workplace. Criteria include flexible work, pay equity and parental leave



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